

CS502-ALGORITHMS
MID –FALL2007

Q#1 Total time for heapify is:

- O ($\log_2 n$)
- O ($n \log n$)
- O ($n^2 \log n$)
- O ($\log n$)

Q#2

Solve the recurrence using iteration method and also find time complexity (Θ notation)

$$T(n) = C + O(1) + T(n-1)$$

$T(1) = 1$ and C is a constant.

Q#3

Suggest the criteria for measuring algorithms. Also discuss the issues need to be discussed in the algorithm design.

Q#4

If an algorithm has a complexity of $\log_2 n + n \log_2 n + n$. we could say that it has complexity

- O(n)
- O($n \log_2 n$)
- O(3)
- O($\log_2 (\log_2 n)$)
- O($\log_2 n$)

Q#5

Let the set $P = \{(1, 13), (2, 9), (3, 15), (4, 12), (5, 14), (6, 6), (7, 3), (8, 10), (9, 2), (10, 8), (11, 9), (13, 6), (15, 3), (18, 5)\}$. You are required to give the final state of stack after the execution of sweep line algorithm for 2d-maxima. No intermediate steps or graphics to be shown.

Q#6

Suppose we have hardware capable of executing 10^6 instructions per second. How long would it take to execute an algorithm whose complexity function is $T(n) = 2n^2$ on an input of size $n = 10^8$?

Q#7

In RAM model instructions are executed
One after another
Parallel

Concurrent

Random

Q#8

In selection algorithm, because we eliminate a constant fraction of the array with each phase, we get the

Convergent geometric series

Divergent geometric series

None of these

Q#9

Due to left-complete nature of binary tree, heaps can be stored in

Link list

Structure

Array

None of above